

Project Design Phase
Proposed Solution Template

Date	14 feb 2026
Team ID	LTVIP2026TMIDS35912
Project Name	Deep Learning Fundus Image Analysis for Early Detection of Diabetic Retinopathy
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1. 1	Problem Statement (Problem to be solved)	Diabetic Retinopathy is a serious eye disease caused by diabetes and is one of the leading reasons for preventable blindness worldwide. In many cases, it progresses without noticeable symptoms until significant vision loss occurs. Traditional screening requires regular eye examinations by specialists using expensive equipment, which is not easily accessible to everyone, especially in rural and underserved areas. Delayed diagnosis leads to irreversible damage, highlighting the urgent need for an early, accurate, and automated detection system.
2. 2	Idea / Solution Description	This project focuses on early detection of Diabetic Retinopathy using deep learning techniques applied to retinal (fundus) images. A Convolutional Neural Network (CNN) model is trained using PyTorch to automatically analyze retinal images and predict whether Diabetic Retinopathy is present or not. The system is integrated into a Django-based web application where users can securely register, upload retinal images, and receive fast, data-driven predictions, enabling timely medical consultation and treatment.
3. 3	Novelty / Uniqueness	1. Uses a non-invasive, image-based approach by analyzing retinal fundus images without manual examination.2. Implements a deep learning CNN model using PyTorch for high-accuracy medical image classification.3. Fully integrated with a Django web application including login, registration, and user profile management.4. Focuses on early-stage detection to prevent vision loss, especially for populations with limited access to ophthalmologists.

4. 4.	Social Impact / Customer Satisfaction	1. Enables early diagnosis and prevention of diabetes-related blindness.2. Reduces dependency on costly and time-consuming eye examinations.3. Provides fast, reliable, and user-friendly screening support for patients and healthcare providers.4. Builds confidence and peace of mind through secure and accurate predictions.5. Highly beneficial for rural and economically weaker communities with limited healthcare infrastructure.
2. 5.	Business Model (Revenue Model)	Freemium Model – Basic screening and predictions are provided free of cost, while advanced features such as detailed reports, specialist consultations, and long-term monitoring can be offered as premium services.
3. 6.	Scalability of the Solution	1. Scalable cloud-based deployment supporting large numbers of users simultaneously.2. Model can be retrained with new datasets to improve accuracy or adapt to different populations.3. Modular system design allows easy extension to detect other eye diseases like glaucoma or cataracts.4. Can be integrated with hospital systems, telemedicine platforms, or mobile health applications.